



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Mid-Autumn Semester Examination 2024-25

Date of Examination: 19th September 2024 Session: AN Duration: 2 Hrs Full Marks: 50

Subject No. : AI61004 Subject : Statistical Foundations of AI/ML Department: Artificial Intelligence

Specific charts, graph paper, log book etc., required _____

Special Instructions (if any) : ALL CALCULATIONS NEED TO BE SHOWN IN FULL DETAILS

Please answer Q1 and any 3 of the remaining Questions. All parts of each question must be answered together.

Q1. A) Define unbiased and consistent estimators. Explain with a synthetic example (no need to consider any actual estimator) how an unbiased estimator need not be consistent and vice versa. **[4+2 = 6 marks]**

Solution Sketch: Clear definitions of both terms are needed with precise mathematical definitions. Both definitions are clearly given in slides 19 and 20 of Module 2, still vast majority could not write these correctly. Regarding the illustration, the basic idea is that in case of unbiased estimator, the estimates obtained from different sample sets may be either more or less than the true value, but the average equals the true value. For consistent estimator, the estimates from different sample sets are close to the true value, but the average need not equal the true value. 2 marks for precise definition of unbiased estimator, 2 marks for precise definition of consistent estimator, 2 more for the illustration. Failure to write mathematical def will result in penalization.

B) What is the Maximum Likelihood Estimator of the Gaussian mean ' μ ' based on ' n ' IID observations following $N(\mu, \sigma^2)$, where ' σ ' is fixed and known? Is it unbiased and consistent?

[2+4= 6 marks]

Solution Sketch: You are expected to derive the MLE by maximizing the likelihood (joint PDF of ' n ' IID observations) from $N(\mu, \sigma^2)$. This MLE will be the sample mean of these

observations, so the corresponding estimator 'TN' will be $(X_1 + X_2 + \dots + X_N)/N$ (where X_1, X_2 , etc are IID random variables following $N(u, s^2)$) [Note the diff. between estimator (RV) and estimate (values)]. To find the bias, we must calculate the expectation of the estimator, which equals the true mean 'u' as $E(X_1) = E(X_2) = \dots = u$. Consistency requires establishing that the estimator is guaranteed to stay close enough to 'u' as N increases. We have already established in class that 'TN' as defined above will itself follow $N(u, s^2/N)$. Probability of the event that $|TN - u| < e = p(u - e < TN < u + e)$ can be calculated as the integration, i.e. the probability mass under the Gaussian curve between 'u-e' and 'u+e'. Now as the variance is s^2/N approaches 0 as N approaches infinity, this probability also increases (since over 99% of the probability mass of the Gaussian lies between $u - 3s/\sqrt{N}$ and $u + 3s/\sqrt{N}$, and this gap becomes tighter with large N). Hence it is a consistent estimator. 1.5 marks for deriving the MLE, 0.5 marks for stating the corresponding estimator, 2 marks for deriving that the bias is 0, and 2 marks for the consistency argument.

C) Give an example of an estimator that is biased, but asymptotically unbiased. Justify with full calculations. **[2+6 = 8 marks]**

Solution Sketch: The intended answer was the MLE of the Gaussian variance (sample variance), but any other valid answer will do, as long as an estimator function is mathematically stated along with the quantity that is being attempted to estimate. 2 marks for defining, and 6 marks for deriving the expression of its bias, along with explanation why that bias should tend to 0 as N approaches infinity.

[General comment: except for the 2 marks regarding consistency of the ML-Estimator, all other parts were provided in the slides/derived clearly in class. So I expected most students to score at least 12 in this question. Yet, over 50% have scored less than 5, and not even understood the meaning of 'unbiased' and 'consistent']

Q2. A) Consider two independent 2D random variables $X \sim N(u_1, \Sigma_1)$ and $Y \sim N(u_2, \Sigma_2)$. Calculate the covariance matrix of $aX + bY$ where a, b are any real numbers. **[5 marks]**

Solution Sketch: Set $Z = aX + bY$, where X, Y, Z are 2D vectors. The definition of covariance matrix is $E((Z-E(Z))*(Z-E(Z))^T)$. This can be calculated by simplifying step by step, using the rules of vector/matrix multiplication, and noting that $E((X-E(X))*(X-E(X))^T) = \Sigma_1$ etc. Directly taking scalar formula related to variance/covariance and using them in vector settings will result in deduction of marks. Also, failure to maintain consistency of vector/matrix operations will be penalized. Moment Generating Function is not a useful approach here since we are looking for matrices.

B) Calculate the mean square error of the maximum likelihood estimator of the Bernoulli and Gaussian mean parameters, based on N IID observations? **[5 marks]**

Solution Sketch: The definition of mean square error is provided in slide 20 of Module 2. There is no need to derive the ML estimators of the mean parameters, stating them is enough. But the expectation involved in MSE has to be evaluated. Those who have used the bias-variance formula, should derive it before use. 1 mark for writing the formulae of the MLEs, 1 more mark for writing the formula of MSE, and 3 marks for evaluating the expectations in both cases.

Q3. A) We have a model $Y \sim N(W^T X, \sigma^2)$ where W, X are real vectors. Further, $W = V \cdot U$, i.e. W is obtained by elementwise multiplication between a real-valued vector V and a random binary vector U such that each element of U follows Bernoulli distribution with parameter ρ . Write down the likelihood function. **[3 marks]**

Solution Sketch: The likelihood function has two parts: the Gaussian for Y , and the Bernoulli for U . In the Gaussian part, $W^T X$ can be written as $(V \cdot U)^T X$ or $\sum_j v_j x_j$, where 'j' indicates the dimensions of the vectors. In the Bernoulli part, it can be written as the product of the Bernoulli PDFs of each of the dimensions. So the likelihood is $1/\sqrt{2\pi\sigma^2} \exp(-(Y - \sum_j v_j x_j)^2 / 2\sigma^2) * \prod_j \rho^{u_j} (1-\rho)^{1-u_j}$ or $1/\sqrt{2\pi\sigma^2} \exp(-(Y - \sum_j v_j x_j)^2 / 2\sigma^2) * \rho^{\sum u_j} (1-\rho)^{\sum (1-u_j)}$

B) We have an observation ($X = [2 \ 2]$, $Y=6$) from the above model. Assuming ($V = [3 \ 3]$, $\sigma^2 = 1$), what is the maximum likelihood estimate of p ? **[3 marks]**

Solution Sketch: Here, we notice that Y , X , V are provided, but not U . Since U is a random variable, we must marginalize over it. There are 4 possibilities for U : (1,1) with probability p^2 , (1,0) or (0,1) with probability $p(1-p)$, and (0,0) with probability $(1-p)^2$. So The above likelihood function can be rewritten accordingly (without u) as only a function of p , and differentiating it w.r.t p and equating to 0 should give the answer.

C) $X_1 \sim \text{Gamma}(a, 1)$, $X_2 \sim \text{Gamma}(b, 1)$. Find the distribution of $Y = X_1/(X_1+X_2)$. **[4 marks]**

Solution Sketch: This is a simplified version of a similar problem involving 3 variables (X_1 , X_2 , X_3) which was worked out for the last set of practice problems. Here we have only X_1 , X_2 , and the target variable Y clearly has support only in (0,1). Like the worked-out problem, the idea is to write the joint distribution of (X_1 , X_2), substitute one of them (say X_2 with Y and X_1), and then marginalize X_1 by integrating for 0 to infinity. The PDF on Y will be of the form $y^{(a-1)}(1-y)^{(b-1)}$, i.e. it will follow Beta distribution.

Q4. A) Show that Dirichlet distribution is a Conjugate Prior for Multinomial Distribution.

[4 marks]

Solution Sketch: Prior $\theta \sim \text{Dir}(\alpha)$, Likelihood $X | \theta \sim \text{Mult}(\theta)$. The posterior $p(\theta | X) \propto p(X | \theta) * p(\theta)$. Put in the PDFs $p(X | \theta)$ and $p(\theta)$, and we get the Dirichlet PDF as their product for posterior.

B) Consider the following model: $Z_i \sim \text{Beta}(a, b)$, $X_i \sim N(WZ_i, s)$. We have 3 observations of X and Z : [$X_1=4.6$ $Z_1=0.2$, $X_2=-2.4$ $Z_2=0.6$, $X_3=1.8$ $Z_3=0.4$]

Consider the following 2 choices of parameters: P1: [$W=5$, $s=9$, $a=4$, $b=4$], P2: [$W=3$, $s=25$, $a=2$, $b=2$]. Which of these two sets of parameters fits the observations better? **[6 marks]**

[Beta(2,2) = 1/6, Beta(4,4) = 1/140]

Solution Sketch: This is actually a very simple problem as it does not involve any marginalization. The likelihood function is the product of the Gaussian (for $X|Z$) and the Beta (for Z), like $\prod_i N(X_i | Z_i) * \text{Beta}(Z_i) = \prod_i (1/\sqrt{2\pi s}) * \exp(-(X_i - W * Z_i)^2 / 2s^2)) * Z_i^{a-1} * (1-Z_i)^{b-1}$. This needs to be calculated separately for both sets of parameters.

Q5. A) Derive the probabilistic version of ridge regression by considering $N(0, \rho^2 I)$ as the prior on the coefficient vector W . **[4 marks]**

Solution Sketch: Slide 35 of Module 2, the posterior PDF is expected to be calculated fully by multiplying the likelihood and prior PDFs.

B) We have 10 data-points of the form $D=\{x_i, y_i\}$ where (x_i, y_i) are both real numbers (given below). I want to fit a model $Y = WX + E$ where W is a real number and E is a Gaussian noise $N(0, 9)$. I have a prior distribution $p(W) = N(5, 1)$ on the parameter W . Derive the expected value for y_{11} , based on x_{11} and the first 10 observations (D). **[6 marks]**

ID	1	2	3	4	5	6	7	8	9	10	11
X	-4	-3	-2	-1	0	1	2	3	4	5	6
Y	-18	-8	-9	0	-3	5	5	11	20	19	?

Solution Sketch: $E(y_{11}) = E(W) * x_{11}$ as the expected noise is 0. $E(W)$ is the mean/mode of the posterior on W calculated from the 10 observations, i.e. $p(W | D) = p(D | W) * p(W)$ where $p(W) = N(W; 5, 1)$ and $p(D | W) = \prod_i N(y_i; W * x_i, 9)$. Write down the posterior PDF as the product of the last two, and calculate $E(W)$ as the maxima of the posterior (obtained by differentiation w.r.t. W). As the posterior is again Gaussian, the mode of the posterior is same as $E(W)$.